# MPG No. 9.3.31 $4'$ [ Type III, tetragonal ] [Q tensor]

* Rank 0 tensor.

$$[Q]$$

$$Q = \mathbb{Q}_0^{(1)}$$

* Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & Q_z \end{bmatrix}$$

$$Q_z = \mathbb{Q}_z^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & 0 \\ 0 & Q_{xx} & 0 \\ 0 & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_u^{(1)}$$
$$Q_{zz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & Q_{xy} & 0 \\ -Q_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xy} = \mathbb{G}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{zzz} \\ 0 & Q_{xxz} & 0 \\ Q_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxz} = \mathbb{Q}_z^{(1)} - \mathbb{Q}_z^{\alpha(1)}$$
$$Q_{zzz} = 3\mathbb{Q}_z^{(1)} + 2\mathbb{Q}_z^{\alpha(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & Q_{xxz} \\ 0 & 0 & Q_{xxz} \\ 0 & 0 & 0 \\ Q_{yzx} & -\frac{Q_{xxz}}{2} & 0 \\ -\frac{Q_{xxz}}{2} & -Q_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxz} = 2\mathbb{Q}_z^{(2)}$$
$$Q_{yzx} = 3\mathbb{G}_u^{(1)}$$
$$\frac{Q_{xxz}}{2} = \mathbb{Q}_z^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} Q_{yzx} & Q_{yzy} & 0 \\ -Q_{yzy} & Q_{yzx} & 0 \\ 0 & 0 & Q_{xyz} \end{bmatrix}$$

$$Q_{yzx} = \mathbb{G}_0^{(1)} - \mathbb{G}_u^{(2)}$$
$$Q_{yzy} = \mathbb{Q}_z^{(3)}$$
$$Q_{xyz} = \mathbb{G}_0^{(1)} + 2\mathbb{G}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & 0 & Q_{xxxy} \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & 0 & 0 & -Q_{xxxy} \\ Q_{xxzz} & Q_{xxzz} & Q_{zzzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzyz} & 0 \\ Q_{xxxy} & -Q_{xxxy} & 0 & 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)}$$
$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)}$$
$$Q_{xxxy} = \mathbb{Q}_{4z}^{\alpha(1)}$$
$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)}$$
$$Q_{xyxy} = \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & Q_{xxzz} & 0 & 0 & Q_{xxxy} \\ 0 & 0 & Q_{xxzz} & 0 & 0 & -Q_{xxxy} \\ -Q_{xxzz} & -Q_{xxzz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & Q_{yzzx} & 0 \\ 0 & 0 & 0 & -Q_{yzzx} & 0 & 0 \\ -Q_{xxxy} & Q_{xxxy} & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxzz} = 3\mathbb{Q}_u^{(3)}$$
$$Q_{xxxy} = 2\mathbb{G}_z^{(1)} - 2\mathbb{G}_z^{\alpha(1)}$$
$$Q_{yzzx} = -\mathbb{G}_z^{(1)} - 4\mathbb{G}_z^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & 0 \\ 0 & Q_{yzyz} & 0 \\ 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)}$$
$$Q_{xyxy} = \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & Q_{yzzx} & 0 \\ -Q_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzzx} = -\mathbb{G}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & Q_{xxxy} \\ 0 & 0 & Q_{xxxy} \\ 0 & 0 & Q_{zzxy} \\ Q_{yzyz} & Q_{yzzx} & 0 \\ Q_{yzzx} & -Q_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$Q_{xxxy} = \mathbb{G}_z^{(3)} - \mathbb{G}_z^{\alpha(2)}$$
$$Q_{zzxy} = \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{(4)} + 2\mathbb{G}_z^{\alpha(2)}$$
$$Q_{yzyz} = -3\mathbb{Q}_u^{(5)}$$
$$Q_{yzzx} = \mathbb{G}_z^{(4)} - \mathbb{G}_z^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & Q_{yzzx} & 0 \\ 0 & 0 & 0 & Q_{yzzx} & -Q_{yzyz} & 0 \\ Q_{xyxx} & Q_{xyxx} & Q_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$Q_{yzyz} = -3\mathbb{Q}_u^{(6)}$$
$$Q_{yzzx} = \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)}$$
$$Q_{xyxx} = \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)}$$
$$Q_{xyzz} = \mathbb{G}_z^{(5)} + 2\mathbb{G}_z^{(6)} + 2\mathbb{G}_z^{\alpha(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & 0 & Q_{xxxy} & 0 & 0 & Q_{xxyx} \\ Q_{xxyy} & Q_{xxxx} & Q_{xxzz} & 0 & 0 & -Q_{xxyx} & 0 & 0 & -Q_{xxxy} \\ Q_{zzxx} & Q_{zzxx} & Q_{zzzz} & 0 & 0 & Q_{zzxy} & 0 & 0 & -Q_{zzxy} \\ 0 & 0 & 0 & Q_{yzyz} & Q_{yzzx} & 0 & Q_{yzzy} & Q_{yzxz} & 0 \\ 0 & 0 & 0 & -Q_{yzxz} & Q_{yzzy} & 0 & -Q_{yzzx} & Q_{yzyz} & 0 \\ Q_{xyxx} & -Q_{xyxx} & 0 & 0 & 0 & Q_{xyxy} & 0 & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxxx} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)}$$
$$Q_{xxyy} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)}$$
$$Q_{xxzz} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)}$$
$$Q_{xxxy} = 2\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)} + \mathbb{Q}_{4z}^{\alpha(1)}$$
$$Q_{xxyx} = 2\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(3)} - 2\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)} + \mathbb{Q}_{4z}^{\alpha(1)}$$
$$Q_{zzxx} = \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)}$$
$$Q_{zzzz} = \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)}$$
$$Q_{zzxy} = \mathbb{G}_z^{(3)} + 2\mathbb{G}_z^{(4)} + 2\mathbb{G}_z^{\alpha(2)}$$
$$Q_{yzyz} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)}$$
$$Q_{yzzx} = -\mathbb{G}_z^{(1)} + \mathbb{G}_z^{(4)} - 4\mathbb{G}_z^{\alpha(1)} - \mathbb{G}_z^{\alpha(2)}$$
$$Q_{yzzy} = \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)}$$
$$Q_{yzxz} = -\mathbb{G}_z^{(1)} - \mathbb{G}_z^{(4)} - 4\mathbb{G}_z^{\alpha(1)} + \mathbb{G}_z^{\alpha(2)}$$
$$Q_{xyxx} = -2\mathbb{G}_z^{(1)} + 2\mathbb{G}_z^{\alpha(1)} + \mathbb{Q}_{4z}^{\alpha(1)}$$
$$Q_{xyxy} = \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & Q_{yzzx} & 0 & Q_{yzzy} & Q_{yzxz} & 0 \\ 0 & 0 & 0 & Q_{yzxz} & -Q_{yzzy} & 0 & Q_{yzzx} & -Q_{yzyz} & 0 \\ Q_{xyyx} & Q_{xyxx} & Q_{xyzz} & 0 & 0 & Q_{xyxy} & 0 & 0 & -Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned}
Q_{yzyz} &= \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} \\
Q_{yzzx} &= -\mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} \\
Q_{yzzy} &= -\mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} \\
Q_{yzxz} &= \mathbb{G}_z^{(2)} + \mathbb{G}_z^{(6)} - \mathbb{G}_z^{\alpha(3)} \\
Q_{xyyx} &= \mathbb{G}_z^{(5)} - \mathbb{G}_z^{\alpha(3)} \\
Q_{xyzz} &= \mathbb{G}_z^{(5)} + 2\mathbb{G}_z^{(6)} + 2\mathbb{G}_z^{\alpha(3)} \\
Q_{xyxy} &= \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxxy} & 0 \\ -Q_{xxxy} & Q_{xxxx} & 0 \\ 0 & 0 & Q_{zzzz} \\ 0 & 0 & Q_{yyzz} \\ Q_{zzxx} & 0 & 0 \\ Q_{xxxy} & Q_{xxyy} & 0 \\ 0 & Q_{yzzy} & 0 \\ 0 & 0 & Q_{zxxz} \\ Q_{xxyy} + Q_{yyzz} - Q_{yzzy} - Q_{zxxz} + Q_{zzxx} & -Q_{xxxy} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
Q_{xxxx} &= 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} \\
Q_{xxxy} &= \mathbb{Q}_{4z}^{\alpha(1)} \\
Q_{zzzz} &= 3\mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)} \\
Q_{yyzz} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} \\
Q_{zzxx} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} \\
Q_{xxyy} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} \\
Q_{yzzy} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} \\
Q_{zxxz} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)}
\end{aligned}$$